COVID 19 PREDICTION BASED ON SYMPTOMS AND NEARBY HOSPITAL APPOINTMENTS SCHEDULING

A CSE 3999 TARP REPORT

submitted by

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in partial fulfilment for the award of the degree of

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in

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DECLARATION BY THE CANDIDATE

We hereby declare that the Industrial Internship report entitled "**COVID 19 PREDICTION BASED ON SYMPTOMS AND NEARBY HOSPITAL APPOINTMENTS SCHEDULING"** submitted by me to Vellore Institute of Technology, Vellore in partial fulfilment of the requirement for the award of the degree of **Bachelor of Technology** in **Computer Science and Engineering** is a project and research undertaken by us under the supervision of **Dr. Archana Tamizharasan.** We further declare that the work reported in this report has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university.



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1. INTRODUCTION

1.1 BACKGROUND

SARS-CoV-2 screening can assist with decreasing the strain on medical services frameworks by taking into consideration an expeditious and precise distinguishing proof of COVID-19. To assess the danger of contamination, expectation models have been fostered that incorporate various elements which had a low precision in distinguishing SARS-CoV-2 without testing. These are planned to help clinical staff all throughout the planet in triaging patients, especially in regions where medical care assets are scant.

We will make a machine learning model that detects COVID-19 cases utilizing basic factors accessible by posing essential inquiries, in view of cross-country information publicly delivered by the Israeli service of Health and furthermore encourage the essential fundamental meds to be taken. Alongside this model we will likewise incorporate forecasts of neighbouring Covid 19 clinics by examining close by emergency clinic areas from the current area by utilizing GPS (Global Positioning System) and Google Maps API (Application Program Interface) and a website utilizing html, CSS, php and MySQL.

Internet Screening can save a great deal of time visiting the medical clinic and hanging tight for the arrangement and there is a lot of hazards of interacting with the tainted individual and increasing the odds of getting tainted as well. The long lines in the medical clinic can take a lot of time without really any assurance of arrangement. However, with internet screening there is no such issue and it can assist individuals with keeping up with social distance and individuals can be protected in their homes and there is no dread of coming in touch with the tainted individual they can get themselves tried through web-based screening with no dread.

Internet screening can set aside traffic issues, fuel and which can help in the ecological discussion since Coronavirus has impacted everybody around and every individual needs to visit the medical clinic.

1.2 OBJECTIVE

This task will empower huge groups to get financial advantages and to expand their benefits through cost reserve funds and arrive at medical clinics on schedule without spending on transport offices. When the framework recognizes the side effects like Covid 19 in individuals,

it will alarm the clients and give counter drugs as an initial treatment measure. This will not just assist them to find out with regards to Covid 19 yet additionally help to control the spread of infection in their body at a beginning stage as it were. On the off chance that we check out this point of view, over the long haul it can save individuals from a ton of difficulties remembering efficient consumption for clinical benefits. They will likewise get saved from the issue of going to the medical clinic, each an ideal opportunity to gather reports and it will save their time and transport use. Not just the application is independent, simple to work and does not require a lot of labour force which reduces expense and gives advantage to the starting organization.

In our task, any of the perspectives of the proposed issue explanation and arrangements does not struggle with legitimate necessities like drafting laws, information insurance acts or web-based media laws which gives it political possibility. Socioeconomics are significant to see how clients look for data, buy items and administrations on the web and the developing need of immense importance so we can foresee the achievement or disappointment of our undertaking ahead of time. The actual area of the clients or customers will likewise decide the prominence of this undertaking because of the speed and inclusion of the web in their space. As indicated by research our item is demographically achievable as well.

2. MOTIVATION

SARS-CoV-2 screening can help reduce the strain on healthcare systems by allowing for a prompt and accurate identification of COVID-19. In order to evaluate the risk of infection, prediction models have been developed that include numerous factors which had a low accuracy in detecting SARS-CoV-2 without testing. These are intended to aid medical personnel around the world in triaging patients, particularly in areas where healthcare resources are scarce. We will create a machine learning model that detects COVID-19 cases using simple variables available by asking basic questions, based on nationwide data publically released by the Israeli Ministry of Health and also advise the necessary basic medicines to be taken. Along with this model we will also integrate prediction of nearby Covid 19 hospitals by scanning nearby hospital locations from the current location by using GPS and Google Maps API and a website using html, css, php and mysql. Online Screening can save a lot of time visiting the hospital and waiting for the appointment and there is too much risk of coming in contact with the infected person and increase the chances of getting infected too. The long queues in the hospital

can take a lot of time with no guarantee of appointment. But with online screening there is no such issue and it can help people maintain social distance and people can be safe in their houses and there is no fear of coming in contact with the infected person and they still can get themselves tested through online screening with no fear. Online screening can save up traffic issues, fuel and which can help in the environmental conversation since covid-19 has affected everyone around and almost every person has to visit the hospital. This project will enable large crowds to get economic benefits and to increase their profits through cost savings and reach hospitals on time without having to spend on transport facilities. As soon as the system identifies the symptoms similar to Covid 19 in people, it will alert the customers and provide counter medicines as an early treatment measure. This will not only help them to get an idea about Covid 19 but also help to curb the spread of virus in their body at an initial stage only. If we look at this perspective, then in the long run it can save people from a lot of troubles including economical expenditure in medical services. They will also get saved from the hassle of travelling to the hospital every time to collect reports and it will save their time and transport usage. Not only this, this application is self sufficient, easy to operate and does not require much workforce which cuts cost and provides benefit to the launching company. In our project, any of the aspects of the proposed problem statement and solutions does not conflict with legal requirements like zoning laws, data protection acts or social media laws which provides it political feasibility. Demographics are important so as to understand how customers search for information, purchase products and services online and the growing need of the hour so that we can predict the success or failure of our project beforehand. The physical location of the customers or consumers will also determine the popularity of this project due to the speed and coverage of the internet in their area. According to research our product is demographically feasible as well.

3. LITERATURE SURVEY

Powerful screening of SARS-CoV-2 empowers fast and proficient determination of COVID-19 and can moderate the weight on medical care frameworks. Expectation models that consolidate a few elements to assess the danger of disease have been created. These plan to help clinical staff worldwide in triaging patients, particularly with regards to restricted medical care assets. We set up an AI (Artificial Intelligence) approach that prepared on records from 51,831

tried people (of whom 4769 were affirmed to have COVID-19). The test set contained information from the ensuing week (47,401 tried people of whom 3624 were affirmed to have COVID-19). Our model anticipated COVID-19 test results with high exactness utilizing just eight parallel elements: sex, age 60 years, known contact with a tainted individual, and the presence of five introductory clinical manifestations. In view of the cross-country information openly announced by the Israeli Ministry of Health, we fostered a model that identifies COVID-19 cases by basic highlights and by posing essential inquiries. Our system can be utilized, among different contemplations, to focus on testing for COVID-19 when testing assets are restricted.[1]A group of 2,618,862 members detailed their expected indications of COVID-19 on a cell phone-based application. Among the 18,401 who had gone through a SARS-CoV-2 test, the extent of members who announced loss of smell and taste was higher in those with a positive test outcome (4,668 of 7,178 people; 65.03%) than in those with an adverse test outcome (2,436 of 11,223 members; 21.71%) (chances ratio=6.74; 95% certainty interval=6.317.21). A model joining manifestations to anticipate plausible contamination was applied to the information from all application clients who announced indications (805,753) and anticipated that 140,312 (17.42%) members are going to have COVID-19.[2] Smell or taste change is a solid indicator for a COVID-19 positive test outcome. Utilizing the presence of smell or taste change with fever, this miserly classifier accurately predicts 75% of COVID-19 test outcomes. A bigger partner of respondents will be important to refine classifier execution.[3]

To further develop wellbeing results, the distinguishing proof and approval of variables that foresee COVID-19 illness movement is essential. Variables including age, comorbidities, resistant reaction, radiographic discoveries, research centre markers, and signs of organ brokenness may independently or anticipate more regrettable results. In any case, the trouble of anticipating COVID-19 sickness seriousness is highlighted by the way that SARS-CoV-2 has tropism for different tissues including fundamentally the respiratory parcel yet in addition the mind, endothelium, heart, kidney, and liver.7 Identification of elements that foresee inconveniences of COVID-19 is urgent for directing clinical consideration, working on tolerant results, and apportioning scant assets.[4 We found that the quantity of effectively evaluated COVID-19 and control cases shifts impressively between indication checkers, with various manifestation checkers showing various qualities as for affectability and particularity. A decent harmony among affectability and explicitness was just accomplished by two indication checkers.[5]

Coronavirus, Coronavirus Disease-2019, has a place with a variety of Coronarias. An infection with no antibody makes unusual devastations in the living souls and monetary and financial frameworks in each country all through the world. It has dubiously stopped everything in the public barbarously. An examination on COVID-19 datasets to comprehend which age bunch is affected because of COVID-19. Unique expectation models are assembled utilizing AI calculations and their exhibitions are registered and assessed. Arbitrary Forest Regressor and Random Forest Classifier beat the other AI models like SVM (Support Vector Machine), KNN+NCA, Decision Tree Classifier, Gaussian Naive Bayesian Classifier, Multilinear Regression, Logistic Relapse and Boost Classifier.[6]

A subset of COVID-19 patients display modified olfactory capacity. Here we dissect mass and single cell RNA-Seq datasets to recognize cell types in the olfactory epithelium and olfactory bulb that express cell passage particles that intercede contamination by SARS-CoV-2 (CoV-2), the causal specialist in COVID-19. We observe that examples from entire olfactory mucosa in species including mouse and human express two key qualities engaged with CoV-2 section, ACE2 and TMPRSS2. Notwithstanding, neither olfactory tangible neurons nor olfactory bulb neurons express these qualities, which are communicated in help cells, undifferentiated organisms, and perivascular cells. These discoveries recommend that CoV-2 contamination of non-neuronal cell types prompts anosmia and related aggravations in smell discernment in COVID-19 patients.

[7] The quick speed of the Covid illness 2019 (COVID-19) pandemic brought about by extreme intense respiratory condition Covid 2 (SARS-CoV-2) presents difficulties to the hearty assortment of populace scale information to address this worldwide wellbeing emergency. They set up the Coronavirus Pandemic Epidemiology (COPE) Consortium to join researchers with ability in large information exploration and the study of disease transmission to foster the COVID Symptom Study, recently known as the COVID Symptom Tracker, versatile application. This application which offers information on hazard factors, prescient side effects, clinical results, and geological hotspots was dispatched in the United Kingdom on 24 March 2020 and the United States on 29 March 2020 and has collected more than 2.8 million clients starting at 2 May 2020. Our drive offers a proof of idea for the repurposing of existing ways to deal with and empower quickly versatile epidemiologic information assortment and investigation, which is basic for an information driven reaction to this general wellbeing challenge.

[8] In December 2019, Covid sickness 2019 (COVID-19), which is brought about by the new Covid extreme intense respiratory condition Covid 2 (SARS-CoV-2), was recognized in Wuhan (Hubei area, China)1; it before long spread across the world. In this continuous pandemic, general wellbeing concerns and the critical requirement for successful remedial measures require a profound comprehension of the study of disease transmission, contagiousness, and pathogenesis of COVID-19. Here we examined clinical, atomic and immunological information from 326 patients with affirmed SARS-CoV-2 disease in Shanghai. The genomic arrangements of SARS-CoV-2, gathered from 112 notable examples along with successions in the Global Initiative on Sharing All Influenza Data (GISAID) dataset, showed a steady development and recommended that there were two significant ancestries with differential openness history during the beginning stage of the episode in Wuhan. In any case, they displayed comparative harmfulness and clinical results. Lymphocytopenia, particularly decreased CD4+ and CD8+ T cell counts upon medical clinic affirmation, was prescient of sickness movement. Significant degrees of interleukin (IL)- 6 and IL-8 during treatment were seen in patients with extreme or basic sickness and associated with diminished lymphocyte count. The determinants of sickness seriousness appeared to stem from factors like age and lymphocytopenia (and its related cytokine storm), while viral hereditary variety did not influence results.

[9] To depict results of individuals conceded to medical clinics with Covid infection 2019 (Coronavirus) in the United States, and the clinical and research facility attributes related with seriousness of ailment. Age and comorbidities were viewed as solid indicators of emergency clinic confirmation and less significantly of basic disease and mortality in individuals with Coronavirus; be that as it may, disability of oxygen on affirmation and markers of irritation were most emphatically connected with basic sickness and mortality. Results appear to be working on over the long run, recommending enhancements in care.[10]

4. OVERVIEW AND WORKING DIAGRAM

4.1 Modules

4.1.1 Module 1

SARS-CoV-2 screening that is effective provides for a fast and reliable diagnosis of COVID-19, for the ease in burden on healthcare systems. Prediction models that integrate numerous

parameters have been developed to determine the likelihood of infection. These are designed to assist medical staff across the world in classifying patients, especially in locations where healthcare resources are scarce or in public spaces.

First, the system will ask the user some basic questions and then compare the answers to the database. If the symptoms match, the patient will be labelled as Covid positive and given a list of medications to take.

4.1.2 Module 2

Using GPS and Google Maps API, this interface allows the patient or anybody else to scan local hospital sites from their present position.

When a user accurately enters his or her local Pin code or name, or utilises GPS, and hits the Find button, the Google Maps API examines the area for a list of suitable hospitals.

When the search is complete, the results are shown in the form of a Dropdown Menu. From that list, the customer may choose their favourite hospital.

4.1.3 Module 3

When a patient clicks on Select and Book Appointment, the server connects them to a website where they may schedule tests or check-ups, book beds, see their planned appointments, and remove them. Doctors may also post reports, and patients can see reports that were previously provided by patients for diagnosis as well as reports that the doctors have uploaded. The platform, on the other hand, accepts lab results, which are updated and saved on the server.

4.2 Working Model

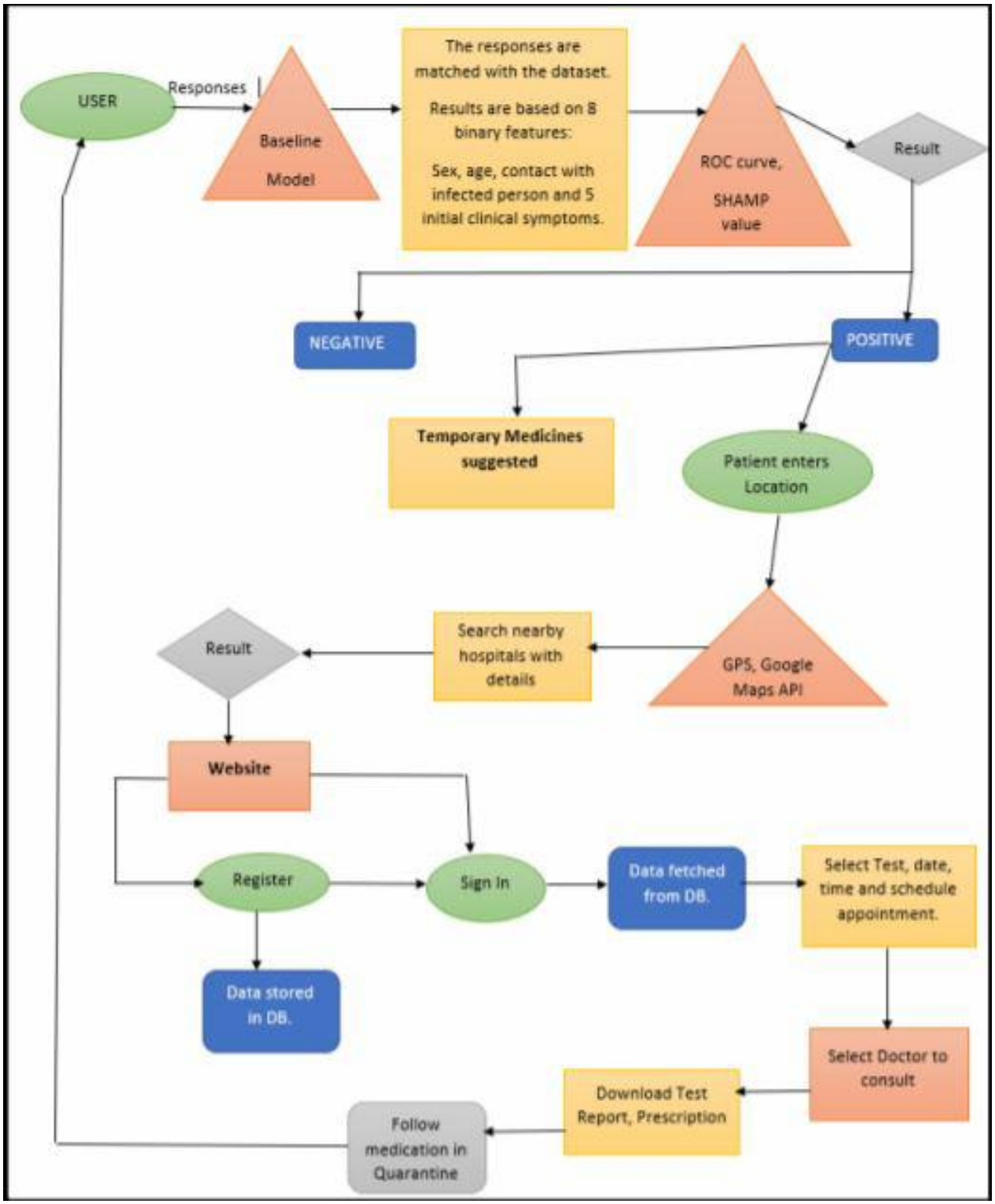


Figure 4.2

5. ALGORITHM AND WORKING

5.1 Phase 1: Building a Prediction model

5.1.1 Logistic Regression

Under the Supervised Learning method, one of the most well-known Machine Learning algorithms is logistic regression. It is used to predict the absolute dependant variable using a set of autonomous factors.

A clear-cut ward variable's yield is predicted using logistic regression. As a result, the outcome should be an all-in or distinct worth. It tends to be either Yes or No, 0 or 1, valid or False, and so on, but instead of presenting the particular worth as 0 or 1, it offers the probabilistic features in the range of 0 to 1.

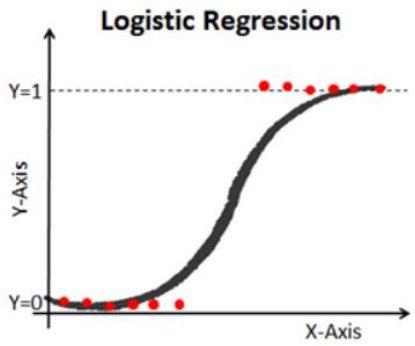
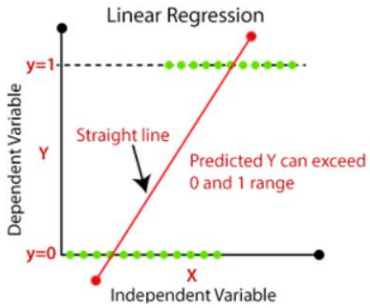


Figure 3.1.1

Except for how they are utilised, logistic regression is identical to linear regression. Straight Regression is used to deal with regression problems, whereas Logistic Relapse is used to deal with characterization problems.

In logistic regression, instead of fitting a relapsing line, we fit a "S" shaped strategic capacity, which predicts the two most severe features (0 or 1). The bend in the logistic function capability shows the likelihood of anything, such as whether the cells are cancerous, whether a mouse is obese or not depending on its weight, and so on.

Logistic Regression is a crucial AI computation since it can provide probabilities and organise current data from both continuous and discrete datasets.Logistic Regression may be used to organise perceptions based on many types of data and can unquestionably determine the optimal criteria to use for the order.



$$y = \beta 0 + \beta 1X1 + \beta 2X2 + \ldots\ldots + \beta nXn$$

Figure 3.1.2

5.1.2 KNN Model

The K Nearest Neighbour method is a type of supervised learning technique that is used for classification and regression. It may also be used to resample datasets and fill in missing values. It predicts the new Datapoint's class or continuous value using K Nearest Neighbours.

The learning process of the algorithm:

1. Instance-based learning: Rather than model-based algorithms, full training examples are used to predict output for the unknown data.

2. Lazy Learning: The model is not learned before the prediction and is postponed until the prediction is asked.

3. Non-Parametric: The mapping function has no pre-set shape in KNN.

KNN finds the distance between the query and all the samples in the data and then selects the defined numbers of samples, called K, nearest to our query. For classification, it decides the most recent label and for regression, it gives the average of labels.

5.1.3 Naïve Bayes Model

Naive Bayes is a machine learning algorithm which is based on probability know as bayes theorem. Bayes theorem helps in calculating conditional probabilities. It helps in solving classification problems which helps in building a machine model to make quick predictions for

binary and multi-class classification problems. It is based on the premise that the predictors are independent. The Naive Bayes classifier assumes that no two features are related to each other.

5.1.4 Building a Prediction Model

The prediction result of the three models' Logistic regression, KNN and Naive Bayes is compared and Naive bayes was found to have higher accuracy and it is considered in the implementation. Naive Bayes algorithm proved to be reliable for a huge amount of data as well as gave faster results than k-NN algorithm. Unlike logistic regression, it works on naive assumption of conditional independence. For trained dataset, Naive Bayes performs better than Logistic Regression and k-NN model in terms of accuracy and time for forecast results. Then we train our dataset and do the model fitting using all the three models and find that using the Naive Bayes model, we get better results in terms of time and accuracy.

It is easy and fast enough to find the data set's class. In comparison to numerical input variables, it gives better results with categorical input variables. We first import the dataset and understand it. Later we analyse the target variable and analyse all the symptoms.When the requirement of independence is met, a Naive Bayes classifier performs better than alternative models and needs less training of data.

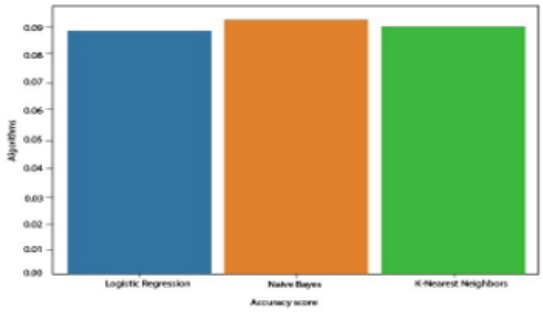


Figure 3.4.1

Model	Accuracy
Logistic regression	89.31 %
KNN	90.07 %
Naïve Bayes	91.13 %

Table :1 – Comparison of Accuracy of the ML Models

5.2 Phase 2 : Developing a Hospital Appointment Scheduler

The World Health Organization has focused on steps to treat a COVID-19 patient starting from finding people and testing if they are affected by the Covid-19 virus, based on the report it is further recommended to isolate the patient and then move ahead with a proper treatment. For the website to display a precise report, it is mandatory for the users to input their gender, age,

and the geolocation for the user to trace information such as number of people already affected in that area and number. Along with personal information the website will then ask the user about some of the major symptoms that are common in Covid-19 patients. Adding to all the information that a user inputs, the person would also be asked if he/she had contacts with other people or recently travelled abroad. This factor is a huge deciding factor of the produced report because there are more chances of being affected if there has been contact with another affected person. For this purpose, a Naive Bayes is run which calculates the total number of people with the age, sex and symptoms are affected by Sars-Cov-2. All the symptoms including fever, headache, shortness of breath, sore throat, loss of smell and taste are prominent signs of being affected by the coronavirus along factors like age and gender which are put together and then compared to the Naive Bayes model and then the report of the user is analysed and an estimated report is printed. Risk of being affected increases if the person already suffers from a chronic disease like heart disease. Therefore, these factors are also taken care of.

If the person is examined as safe from coronavirus it will be displayed that the person is safe but it is still recommended to go for a professional check-up. Even if no such symptoms are found, the user would still be advised to take a non-symptomatic test for Covid-19 and can schedule the next assessment when it is suitable because a negative test does not mean that a person is safe. A person may have an asymptomatic disease. In case the person is detected to be prone to Sars-Cov-2, the user will then be provided with recommended medicines for the symptoms and be given an option to search nearby hospitals. The scan for hospitals is made possible with the help of Google maps using the current location of the user as it is the fastest method to locate hospitals in the vicinity. Using the pin code of the user this website, with the help of GPS and Google API, will locate nearest hospitals with all the facilities available for a patient including hospital beds and oxygen cylinders for worse cases. The user can book an appointment with the doctor by signing up in the website which would create their profile maintaining their health history with all the test reports, appointments, and recovery report. All the reports would be maintained in the database and can even be downloaded from the site. The user is declared as recovered when he/she receives a negative Covid-19 report. Users can revisit the site on a regular basis to keep themselves updated. Hey can schedule the test according to their suitable time preference and get proper guidance about their health and steps for the prevention of Coronavirus.

5.3 Flowchart

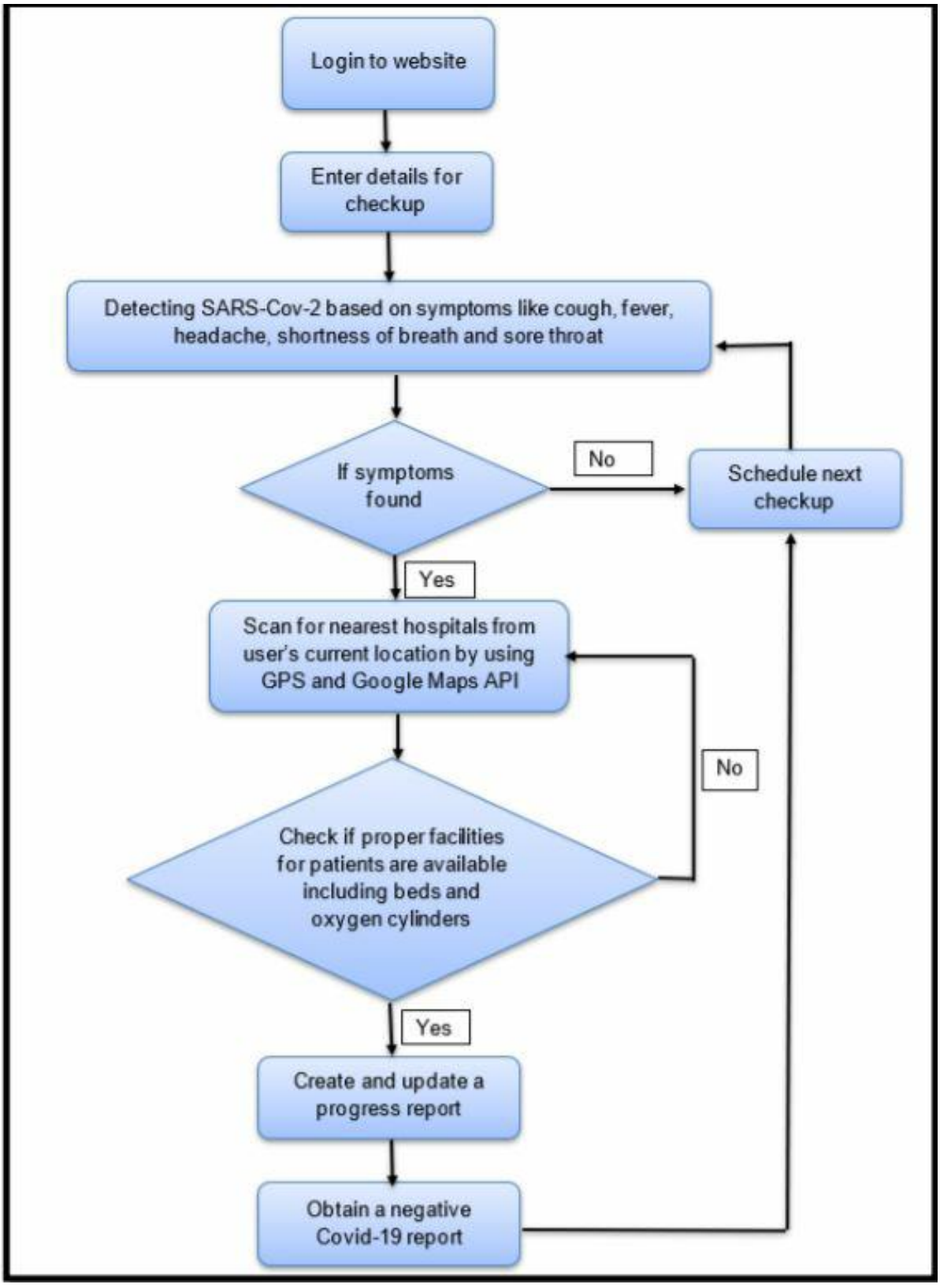


Figure 5.2

6. CODE AND IMPLEMENTATION

6.1 Code

6.1.1 Main.py

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from flask import *
import tensorflow as tf
app = Flask(__name__)


app.jinja_env.filters['zip'] = zip


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'People tree hospitals'

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'Lakeshore hospital kochi',

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],

7:[

'https://patientportal.rajagirihospital.com/',

'https://www.vpslakeshorehospital.com/appointment',

'https://www.one-aster.com/webconnect/#/bookAppointment',

'https://lisiehospital.org/advance-booking#verticalTab3',

'http://www.emccochin.com/doctor-appointment-booking/'

],

8:[

'https://rbh.ckbirlahospitals.com/get-appointment/',

'https://www.medipulse.in/book-an-appointment',

'https://www.sonihospitals.com/appointment/',

'http://essh.in/doctors?view=appointments&specialist',

'https://monilek.onlineappointment.in/'

],

9:[

'https://nemcarehospital.in/info/Patient%20&%20Visitor/Outpatient%20Services',

'https://www.aryahospital.com/',

'http://internationalhosp.com/contact.php',

'https://downtownhospitals.in/#gsc.tab=0'

],

10:[

'http://www.sgpgims.in/pgi/onlineRegHeader.jsf',

'https://www.saharahospitals.com/send-your-details',

'https://www.medantaeclinic.org/#!/home?utm_source=medanta.org&utm_medium=organic&utm_campaign=appointment',

'https://midlandhealthcare.org/book-an-appointment',

'https://www.shubhamhospitals.com/OPD-Schedule/home',

'https://jeevanjyotihospital.com/opd_consultant/',

'https://pritihospital.com/contact_us.html',

'https://www.shakuntalahospital.org/contact.php'

],

11:[

'https://www.medantaeclinic.org/#!/home?utm_source=medanta.org&utm_medium=organic&utm_campaign=appointment',

'https://www.maxhealthcare.in/book-an-appointment',

'https://www.artemishospitals.com/make-an-apointment',

'http://w-hospital.in/book-appointment',

'https://www.cloudninecare.com/registration-and-appointments'

],

12:[

'https://kd.doctor9.com/default.htm',

'https://www.shalby.org/contact-us/sg-shalby-hospital/',

'https://www.cims.org/',

'https://www.sterlinghospitals.com/find-a-doctor/',

'https://zydushospitals.com/Anand/book_appointment.html',

'https://www.hcghospitals.in/BookService'

],

13:[

'https://narayana.health/phrportal',

'https://www.rameswaranursinghome.com/book-a-bed'

],

14:[

'http://app.kalingahospital.com/kalingahospital/Doctor',

'https://vivekanandhospital.com/page/doctor',

'https://www.amrihospitals.in/bhubaneswar#hero-form',

'https://sumum.soahospitals.com/contact-us',

'https://appointments.carehospitals.com/#/home'

],

15:[

'https://bansalhospital.com/',

'https://www.shalby.org/specialities/oncology/indore/',

'https://www.lifecarehospital.co.in/',

```
        'https://www.galaxyhospitalbhopal.com/timing.html',
        'https://pushpanjalihospital.co.in/admission-process'
    ],

    16:[
        'https://www.parashospitals.com/request-an-appointment/',
        'https://mediparkhospitals.com/contact/',
        'http://www.shrinivashospital.com/(S(rlo1muf0iuukgn55veget2jo))/Appointment.aspx',
        'https://rubanpatliputrahospital.com/appointment/opd.php'
    ],

    17:[
        'https://www.fortishealthcare.com/health-packages'
    ]

}

def init():
    with open('model_training.json','r') as f:
        model=tf.keras.models.model_from_json(f.read())
    model.load_weights('model_training.h5')
    return model

model=init()

@app.route('/',methods=['GET','POST'])
def form():
```

```
    if request.method == 'GET':
        return render_template('checkupform.html')
    if request.method == 'POST':
        cough=request.form.get('cough')
        fever=request.form.get('fever')
        sthroat=request.form.get('sthroat')
        reason=request.form.get('reason')
        sob=request.form.get('sob')
        age60=request.form.get('age60')
        headache=request.form.get('headache')
        gender=request.form.get('gender')

data=[[int(cough),int(fever),int(sthroat),int(sob),int(headache),int(age60),int(gender),int(reaso
n)]]
        res=round(model.predict(data)[0][0]*100,2)
        return render_template('index.html',value=res)


@app.route('/hospitals',methods=['GET','POST'])
def hospitals():
    if request.method == 'GET':
        return render_template('hospitals.html')
    if request.method == 'POST':
        state=int(request.form.get('states'))
        h_name=h_names[state]
        h_link=h_links[state]
        return render_template('hospitallist.html',value1=h_name,value2=h_link)


@app.route('/medicines',methods=['GET','POST'])
```

```
def medicines():
    if request.method == 'GET':
        return render_template('medicines.html')
    if request.method == 'POST':
        return render_template('medicines.html')



@app.errorhandler(500)
def error(e):
    return render_template("500.html")

@app.errorhandler(404)
def error(e):
    return render_template("404.html")

if __name__ == "__main__":
    app.run(debug=True, port=5000)
```

6.1.2 index.html

```
<!DOCTYPE html>
<html lang="en">

<head>
        <meta charset="utf-8">
        <meta http-equiv="X-UA-Compatible" content="IE=edge">
        <meta name="viewport" content="width=device-width, initial-scale=1">
```

```
        <title>Prediction</title>


        <!-- Google font -->
        <link href="https://fonts.googleapis.com/css?family=Montserrat:700,900" rel="stylesheet">


        <!-- Font Awesome Icon -->
        <link type="text/css" rel="stylesheet" href="static/css/font-awesome.min.css" />


        <!-- Custom stlylesheet -->
        <link type="text/css" rel="stylesheet" href="static/css/style.css" />



</head>


<body>


        <div id="notfound" style="background-image: url('static/images/bg.jpg');">
                <div class="notfound-bg"></div>
                <div class="notfound">
                        <div class="notfound-404">
                                <h1>{{value}}%</h1>
                        </div>
                        <h2>is the chances of being positive. Please choose below option if the probability is greater than 70</h2>
                        <a href="/medicines" class="home-btn">Prescribed Medicines</a>
                        <a href="/hospitals" class="contact-btn">Nearby Hospitals</a>
```

```
            </div>
        </div>

</body>

</html>
```

6.1.3 checkupforum.html

```
<!doctype html>
<html lang="en">
  <head>
        <title>PRESCRIBED MEDICINES</title>
    <meta charset="utf-8">
    <meta name="viewport" content="width=device-width, initial-scale=1, shrink-to-fit=no">

        <link href='https://fonts.googleapis.com/css?family=Roboto:400,100,300,700'
rel='stylesheet' type='text/css'>

        <link rel="stylesheet" href="https://stackpath.bootstrapcdn.com/font-
awesome/4.7.0/css/font-awesome.min.css">

        <link rel="stylesheet" href="static/css/bootstrap.min.css" type='text/css'>
        <link rel="stylesheet" href="static/css/style1.css" type='text/css'>

        </head>
        <body>
```

```
<section class="ftco-section">
    <div class="container">
        <div class="row justify-content-center">
            <div class="col-md-6 text-center mb-4">
                <h2 class="heading-section">PRESCRIBED MEDICINES FOR COVID</h2>
            </div>
        </div>
        <div class="row">
            <div class="col-md-12">
                <div class="table-wrap">
                    <table class="table myaccordion table-hover" id="accordion">
                      <thead>
                        <tr>
                          <th>#</th>
                          <th>Medicine Name</th>
                          <th>Price</th>
                          <th>Quantity</th>
                          <th> </th>
                        </tr>
                      </thead>
                      <tbody>
                        <tr data-toggle="collapse" data-target="#collapseOne" aria-expanded="true" aria-controls="collapseOne">
                          <th scope="row">1</th>
                          <td>Paracetamol</td>
                          <td>Rs 30.00</td>
```

<td>15 tablets</td>

<td>

</td>

</tr>

<tr>

<td colspan="6" id="collapseOne" class="collapse show acc" data-parent="#accordion">

The use of paracetamol to reduce fever should be considered particularly safe, if mild COVID-19 has not yet been diagnosed, at least in the professional intention of the majority of physicians.

</td>

</tr>

<tr data-toggle="collapse" data-target="#collapseTwo" aria-expanded="false" aria-controls="collapseTwo" class="collapsed">

<th scope="row">2</th>

<td>Budesonide</td>

<td>Rs 27.00</td>

<td>2 ml</td>

<td>

</td>

</tr>

<tr>

<td colspan="6" id="collapseTwo" class="collapse acc" data-parent="#accordion">

When administered in early COVID-19, the inhaled glucocorticoid budesonide may reduce the necessity for urgent medical care and may reduce the time to recovery.

```
                    </td>
                </tr>

                <tr data-toggle="collapse" data-target="#collapseThree" aria-expanded="false" aria-controls="collapseThree" class="collapsed">
                    <th scope="row">3</th>
                    <td>Remdesivir</td>
                    <td>Rs 2500.00</td>
                    <td>100 mg</td>
                    <td>
                        <i class="fa" aria-hidden="false"></i>
                    </td>
                </tr>
                    <tr>
                        <td colspan="6" id="collapseThree" class="collapse acc" data-parent="#accordion">
                            <p>WHO has issued a conditional recommendation against the use of remdesivir in hospitalized patients, regardless of disease severity, as there is currently no evidence that remdesivir improves survival and other outcomes in these patients.</p>
                        </td>
                    </tr>

                <tr data-toggle="collapse" data-target="#collapseFour" aria-expanded="false" aria-controls="collapseFour" class="collapsed">
                    <th scope="row">4</th>
                    <td>Vitamin C tablets</td>
                    <td>Rs 99.00</td>
                    <td>30 tablets</td>
```

<td>

</td>

</tr>

<tr>

<td colspan="6" id="collapseFour" class="collapse acc" data-parent="#accordion">

Vitamin C is perhaps the most popular supplement taken to protect against infection due to its important role in immune health.

This vitamin supports the function of various immune cells and enhances their ability to protect against infection. It’s also necessary for cellular death, which helps keep your immune system healthy by clearing out old cells and replacing them with new ones.

</td>

</tr>

<tr data-toggle="collapse" data-target="#collapseFive" aria-expanded="false" aria-controls="collapseFive" class="collapsed">

<th scope="row">5</th>

<td>Vitamin D tablets</td>

<td>Rs 146.00</td>

<td>30 tablets</td>

<td>

</td>

</tr>

<tr>

<td colspan="6" id="collapseFive" class="collapse acc" data-parent="#accordion">

Vitamin D is a fat-soluble nutrient essential to the health and functioning of your immune system.

Vitamin D enhances the pathogen-fighting effects of monocytes and macrophages — white blood cells that are important parts of your immune defense — and decreases inflammation, which helps promote immune response.

</td>

</tr>

<tr data-toggle="collapse" data-target="#collapseSix" aria-expanded="false" aria-controls="collapseSix" class="collapsed">

<th scope="row">6</th>

<td>Zinc tablets</td>

<td>Rs 149.00</td>

<td>30 tablets</td>

<td>

</td>

</tr>

<tr>

<td colspan="6" id="collapseSix" class="collapse acc" data-parent="#accordion">

Zinc is a mineral that’s commonly added to supplements and other healthcare products like lozenges that are meant to boost your immune system. This is because zinc is essential for immune system function.

</td>

</tr>

```
                                        </tbody>
                                    </table>
                                </div>
                            </div>
                        </div>
                    </div>
                </section>

                <script src="static/js/jquery.min.js"></script>
  <script src="static/js/popper.js"></script>
  <script src="static/js/bootstrap.min.js"></script>
  <script src="static/js/main.js"></script>

            </body>
</html>
```

6.1.4 medicines.html

```
<!doctype html>
<html lang="en">
  <head>
        <title>PRESCRIBED MEDICINES</title>
    <meta charset="utf-8">
    <meta name="viewport" content="width=device-width, initial-scale=1, shrink-to-fit=no">

        <link href='https://fonts.googleapis.com/css?family=Roboto:400,100,300,700'
rel='stylesheet' type='text/css'>
```

```
        <link rel="stylesheet" href="https://stackpath.bootstrapcdn.com/font-
awesome/4.7.0/css/font-awesome.min.css">


        <link rel="stylesheet" href="static/css/bootstrap.min.css" type='text/css'>

        <link rel="stylesheet" href="static/css/style1.css" type='text/css'>


        </head>

        <body>

        <section class="ftco-section">

                <div class="container">

                        <div class="row justify-content-center">

                                <div class="col-md-6 text-center mb-4">

                                        <h2 class="heading-section">PRESCRIBED
MEDICINES FOR COVID</h2>

                                </div>

                        </div>

                        <div class="row">

                                <div class="col-md-12">

                                        <div class="table-wrap">

                                                <table class="table myaccordion table-hover"
id="accordion">

                                                  <thead>

                                                    <tr>

                                                      <th>#</th>

                                                      <th>Medicine Name</th>

                                                      <th>Price</th>

                                                      <th>Quantity</th>

                                                      <th> </th>
```

```
                    </tr>
                </thead>
                <tbody>
                    <tr data-toggle="collapse" data-target="#collapseOne" aria-expanded="true" aria-controls="collapseOne">
                        <th scope="row">1</th>
                        <td>Paracetamol</td>
                        <td>Rs 30.00</td>
                        <td>15 tablets</td>
                        <td>
                            <i class="fa" aria-hidden="true"></i>
                        </td>
                    </tr>
                    <tr>
                        <td colspan="6" id="collapseOne" class="collapse show acc" data-parent="#accordion">
                            <p>The use of paracetamol to reduce fever should be considered particularly safe, if mild COVID-19 has not yet been diagnosed, at least in the professional intention of the majority of physicians.</p>
                        </td>
                    </tr>

                    <tr data-toggle="collapse" data-target="#collapseTwo" aria-expanded="false" aria-controls="collapseTwo" class="collapsed">
                        <th scope="row">2</th>
                        <td>Budesonide</td>
                        <td>Rs 27.00</td>
                        <td>2 ml</td>
                        <td>
```

</td>

</tr>

<tr>

<td colspan="6" id="collapseTwo" class="collapse acc" data-parent="#accordion">

When administered in early COVID-19, the inhaled glucocorticoid budesonide may reduce the necessity for urgent medical care and may reduce the time to recovery.

</td>

</tr>

<tr data-toggle="collapse" data-target="#collapseThree" aria-expanded="false" aria-controls="collapseThree" class="collapsed">

<th scope="row">3</th>

<td>Remdesivir</td>

<td>Rs 2500.00</td>

<td>100 mg</td>

<td>

</td>

</tr>

<tr>

<td colspan="6" id="collapseThree" class="collapse acc" data-parent="#accordion">

WHO has issued a conditional recommendation against the use of remdesivir in hospitalized patients, regardless of disease severity, as there is currently no evidence that remdesivir improves survival and other outcomes in these patients.

</td>

```
                                                </tr>


                                                <tr data-toggle="collapse" data-
target="#collapseFour" aria-expanded="false" aria-controls="collapseFour"
class="collapsed">

                                                  <th scope="row">4</th>

                                                  <td>Vitamin C tablets</td>

                                                  <td>Rs 99.00</td>

                                                  <td>30 tablets</td>

                                                  <td>

                                                      <i class="fa" aria-hidden="false"></i>

                                            </td>

                                                </tr>

                                                        <tr>

                                                    <td colspan="6" id="collapseFour"
class="collapse acc" data-parent="#accordion">

                                                                <p>Vitamin C is perhaps the most
popular supplement taken to protect against infection due to its important role in immune
health.

                                                                This vitamin
supports the function of various immune cells and enhances their ability to protect against
infection. It’s also necessary for cellular death, which helps keep your immune system healthy
by clearing out old cells and replacing them with new ones.</p>

                                                    </td>

                                                </tr>


                                                    <tr data-toggle="collapse" data-
target="#collapseFive" aria-expanded="false" aria-controls="collapseFive"
class="collapsed">

                                                        <th scope="row">5</th>

                                                        <td>Vitamin D tablets</td>
```

<td>Rs 146.00</td>

<td>30 tablets</td>

<td>

</td>

</tr>

<tr>

<td colspan="6" id="collapseFive" class="collapse acc" data-parent="#accordion">

Vitamin D is a fat-soluble nutrient essential to the health and functioning of your immune system.

Vitamin D enhances the pathogen-fighting effects of monocytes and macrophages — white blood cells that are important parts of your immune defense — and decreases inflammation, which helps promote immune response.

</td>

</tr>

<tr data-toggle="collapse" data-target="#collapseSix" aria-expanded="false" aria-controls="collapseSix" class="collapsed">

<th scope="row">6</th>

<td>Zinc tablets</td>

<td>Rs 149.00</td>

<td>30 tablets</td>

<td>

</td>

</tr>

<tr>

```
                                                    <td colspan="6"
id="collapseSix" class="collapse acc" data-parent="#accordion">

                                                        <p>Zinc is a mineral
that’s commonly added to supplements and other healthcare products like lozenges that are
meant to boost your immune system. This is because zinc is essential for immune system
function.</p>

                                                    </td>

                                                </tr>


                                            </tbody>

                                        </table>

                                    </div>

                                </div>

                            </div>

                        </div>

                    </section>


                    <script src="static/js/jquery.min.js"></script>

  <script src="static/js/popper.js"></script>

  <script src="static/js/bootstrap.min.js"></script>

  <script src="static/js/main.js"></script>


                    </body>

</html>
```

6.1.5 hospitals.html

```
<!DOCTYPE html>

<html lang="en">
```

```
 <head>

  <meta charset="utf-8">
  <meta name="viewport" content="width=device-width, initial-scale=1, shrink-to-fit=no">
  <meta name="description" content="">
  <meta name="author" content="">

  <title>States</title>



  <!-- Additional CSS Files -->
  <link rel="stylesheet" type="text/css" href="static/css/bootstrap.min.css">
  <link rel="stylesheet" type="text/css" href="static/css/main.css">



  <link rel="stylesheet" href="static/css/states.css">

  </head>

  <body>

<section class="mini" id="work-process">
      <div class="mini-content">
        <div class="container">
```

```
        <form action='/hospitals' method="post">
<!-- ***** Mini Box Start ***** -->
            <div class="row">
              <div class="col-lg-2 col-md-3 col-sm-6 col-6">
                <a href="#" class="mini-box">
                  <i><img src="static/images/work-process-item-01.png" alt=""></i>
                  <input type="radio" id="states" name="states" value="1">
                  <strong>Delhi</strong>
                </a>
              </div>

              <div class="col-lg-2 col-md-3 col-sm-6 col-6">
                <a href="#" class="mini-box">
                  <i><img src="static/images/work-process-item-01.png" alt=""></i>
                  <input type="radio" id="states" name="states" value="2">
                  <strong>Punjab</strong>
                </a>
              </div>

              <div class="col-lg-2 col-md-3 col-sm-6 col-6">
                <a href="#" class="mini-box">
                  <i><img src="static/images/work-process-item-01.png" alt=""></i>
                  <input type="radio" id="states" name="states" value="3">
                  <strong>Maharashtra</strong>
                </a>
              </div>
```

```
    <div class="col-lg-2 col-md-3 col-sm-6 col-6">
      <a href="#" class="mini-box">
        <i><img src="static/images/work-process-item-01.png" alt=""></i>
        <input type="radio" id="states" name="states" value="4">
        <strong>West Bengal</strong>
      </a>
    </div>

    <div class="col-lg-2 col-md-3 col-sm-6 col-6">
      <a href="#" class="mini-box">
        <i><img src="static/images/work-process-item-01.png" alt=""></i>
        <input type="radio" id="states" name="states" value="5">
        <strong>Tamil Nadu</strong>
      </a>
    </div>

    <div class="col-lg-2 col-md-3 col-sm-6 col-6">
      <a href="#" class="mini-box">
        <i><img src="static/images/work-process-item-01.png" alt=""></i>
        <input type="radio" id="states" name="states" value="6">
        <strong>Karnataka</strong>
      </a>
    </div>
  </div>
```

```
<div class="row">
    <div class="col-lg-2 col-md-3 col-sm-6 col-6">
      <a href="#" class="mini-box">
        <i><img src="static/images/work-process-item-01.png" alt=""></i>
        <input type="radio" id="states" name="states" value="7">
        <strong>Kerala</strong>
      </a>
    </div>

    <div class="col-lg-2 col-md-3 col-sm-6 col-6">
      <a href="#" class="mini-box">
        <i><img src="static/images/work-process-item-01.png" alt=""></i>
        <input type="radio" id="states" name="states" value="8">
        <strong>Rajasthan</strong>
      </a>
    </div>

    <div class="col-lg-2 col-md-3 col-sm-6 col-6">
      <a href="#" class="mini-box">
        <i><img src="static/images/work-process-item-01.png" alt=""></i>
        <input type="radio" id="states" name="states" value="9">
        <strong>Assam</strong>
      </a>
    </div>

    <div class="col-lg-2 col-md-3 col-sm-6 col-6">
      <a href="#" class="mini-box">
```

```
            <i><img src="static/images/work-process-item-01.png" alt=""></i>
            <input type="radio" id="states" name="states" value="10">
            <strong>Utttar Pradesh</strong>
        </a>
    </div>

        <div class="col-lg-2 col-md-3 col-sm-6 col-6">
        <a href="#" class="mini-box">
            <i><img src="static/images/work-process-item-01.png" alt=""></i>
            <input type="radio" id="states" name="states" value="11">
            <strong>Haryana</strong>
        </a>
      </div>

      <div class="col-lg-2 col-md-3 col-sm-6 col-6">
        <a href="#" class="mini-box">
            <i><img src="static/images/work-process-item-01.png" alt=""></i>
            <input type="radio" id="states" name="states" value="12">
            <strong>Gujarat</strong>
        </a>
      </div>
</div>

<div class="row">

        <div class="col-lg-2 col-md-3 col-sm-6 col-6">
        <a href="#" class="mini-box">
```

```
        <i><img src="static/images/work-process-item-01.png" alt=""></i>
              <input type="radio" id="states" name="states" value="13">
        <strong>Jammu and Kashmir</strong>
    </a>
</div>


<div class="col-lg-2 col-md-3 col-sm-6 col-6">
    <a href="#" class="mini-box">
        <i><img src="static/images/work-process-item-01.png" alt=""></i>
        <input type="radio" id="states" name="states" value="14">
        <strong>Orrisa</strong>
    </a>
</div>

    <div class="col-lg-2 col-md-3 col-sm-6 col-6">
    <a href="#" class="mini-box">
        <i><img src="static/images/work-process-item-01.png" alt=""></i>
        <input type="radio" id="states" name="states" value="15">
        <strong>Madhya Pradesh</strong>
    </a>
</div>

<div class="col-lg-2 col-md-3 col-sm-6 col-6">
    <a href="#" class="mini-box">
        <i><img src="static/images/work-process-item-01.png" alt=""></i>
        <input type="radio" id="states" name="states" value="16">
```

```
                <strong>Bihar</strong>
              </a>
            </div>

            <div class="col-lg-2 col-md-3 col-sm-6 col-6">
              <a href="#" class="mini-box">
                <i><img src="static/images/work-process-item-01.png" alt=""></i>
                <input type="radio" id="states" name="states" value="17">
                <strong>Himachal pradesh</strong>
              </a>
            </div>
        </div>

        <div >
          <button class="login100-form-btn" name="submit" input type='submit'>
            Submit
          </button>
        </div>


    </form>

</div>
</div>
</section>
```

```
            <!-- ***** Mini Box End ***** -->

</body>
</html>
```

6.1.6 hospitallist.html

```
<!DOCTYPE html>
<html lang="en">

  <head>

    <meta charset="utf-8">
    <meta name="viewport" content="width=device-width, initial-scale=1, shrink-to-fit=no">
    <meta name="description" content="">
    <meta name="author" content="">

    <title>Hospital List</title>

    <!-- Additional CSS Files -->
    <link rel="stylesheet" type="text/css" href="static/css/bootstrap.min.css">


    <link rel="stylesheet" href="static/css/style2.css">

    </head>

    <body>
```

```
    <section class="section colored" id="pricing-plans">
  <div class="container">
    <!-- ***** Section Title Start ***** -->
    <div class="row">
      <div class="col-lg-12">
        <div class="center-heading">
          <h2 class="section-title">Hospital list</h2>
        </div>
      </div>
      <div class="offset-lg-3 col-lg-6">
        <div class="center-text">
          <p>Please select the nearest hospital to book the appointment</p>
        </div>
      </div>
    </div>
    <!-- ***** Section Title End ***** -->


    <div class="row">

                {% for result,resul2 in value1|zip(value2) %}
      <div class="col-lg-4 col-md-6 col-sm-12" data-scroll-reveal="enter bottom move 50px over 0.6s after 0.2s">
        <div class="pricing-item">
          <div class="pricing-header">
            <h3 class="pricing-title">{{result}}</h3>
```

```
                </div>
                <div class="pricing-footer">
                    <a href= {{resul2}} class="main-button">Book Appointment</a>
                </div>
              </div>
            </div>

                        {% endfor %}


    </div>
</section>

        </body>
        </html>
```

6.2 Implementation of Website

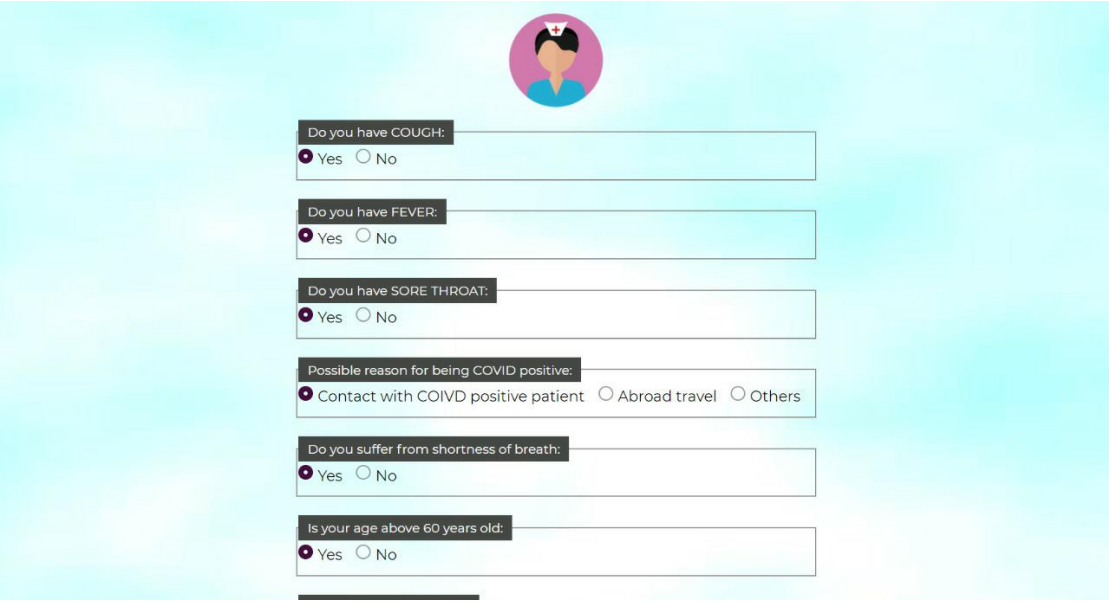


Figure 5.4.1: Panel to enter symptoms

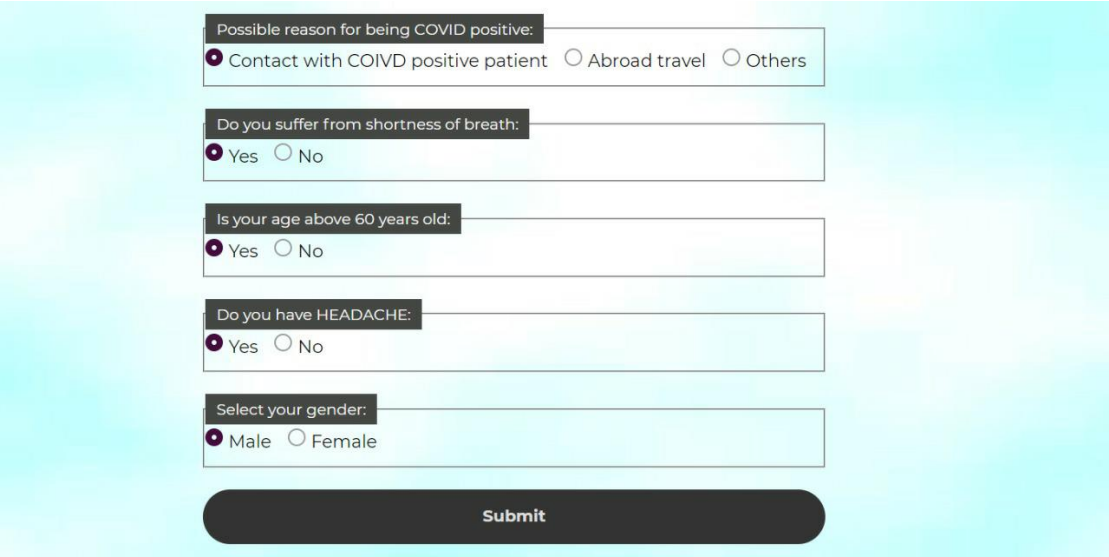


Figure 5.4.2: Panel to submit user details

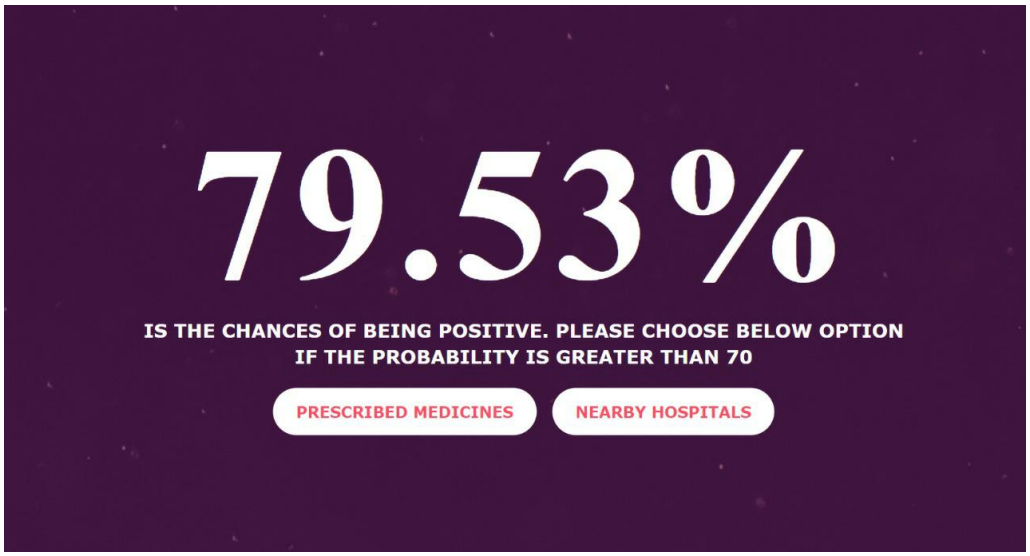


Figure 5.4.3: Result

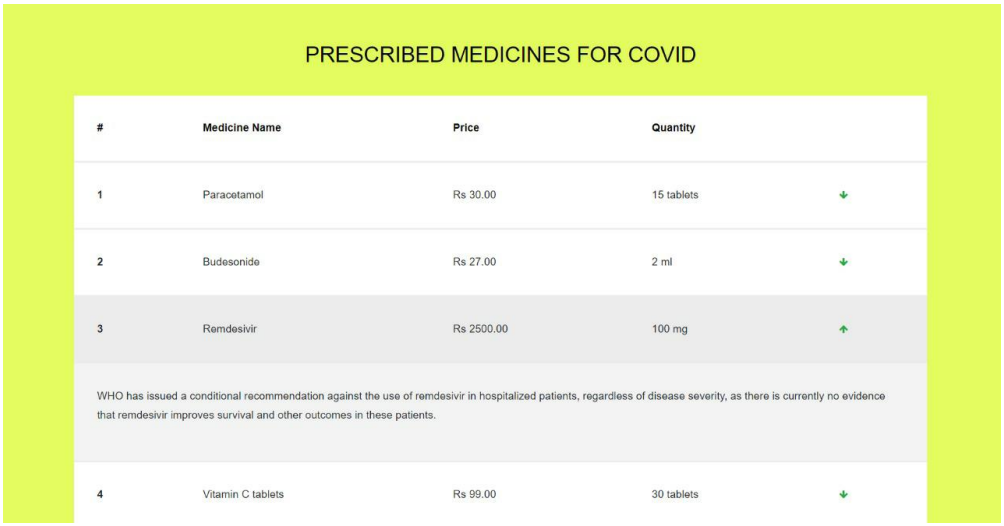


Figure 5.4.4: Prescribed medicines

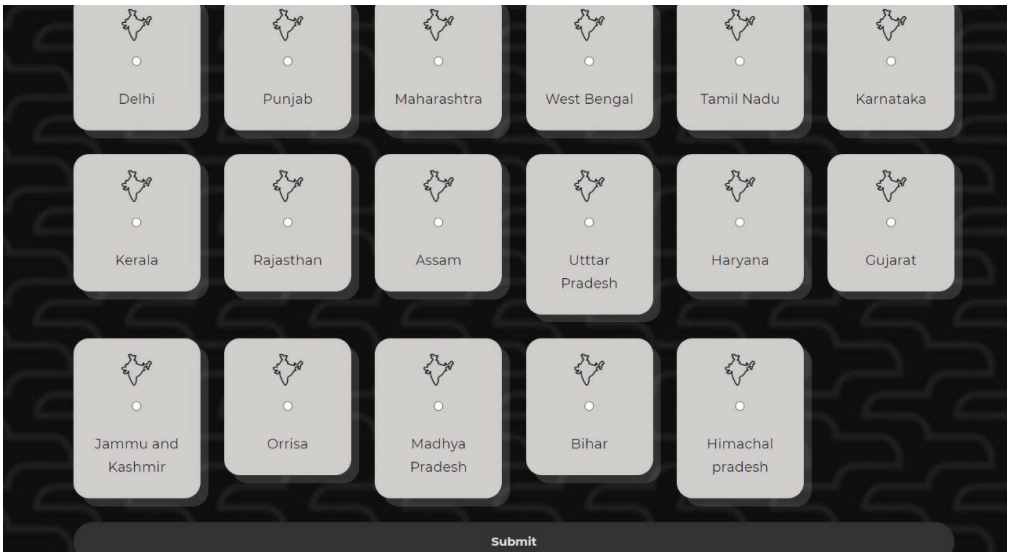


Figure 5.4.5: Window to select state to search hospitals

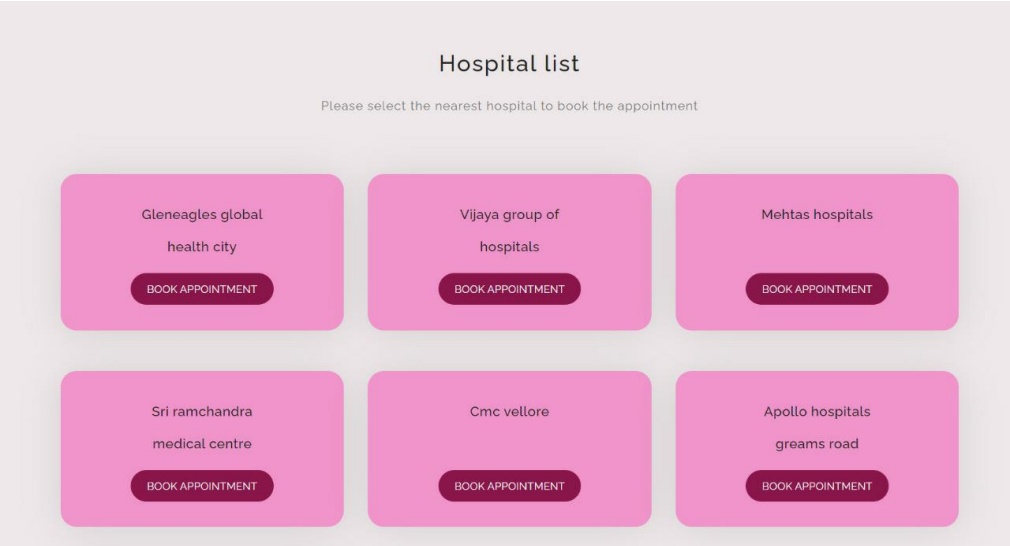


Figure 5.4.6: Window to select a favourable hospital

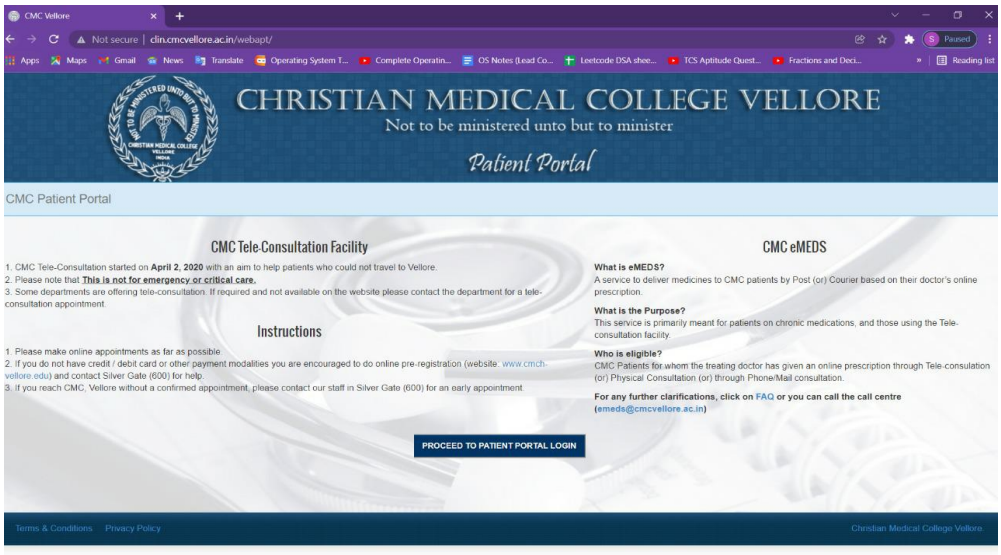


Figure 5.4.7: Redirection to official site of hospital

7. CONCLUSION

Among the models chosen for study such as Logistic regression, KNN and Naive Bayes, Naive bayes was found to have higher accuracy and implemented in building the website. It is easy and fast enough to find the data set's class. When the requirement of independence is met, a Naive Bayes classifier performs better than alternative models and needs less training of data. In comparison to numerical input variables, it gives better results with categorical input variables. We first import the dataset and understand it. Later we analyse the target variable and analyse all the symptoms. Then we train our dataset and do the model fitting using all the three

models and find that using the Naive Bayes model, we get better results in terms of time and accuracy.

In this work, a self-reliant website is constructed to assist the citizens in the times of Corona pandemic. With Sars-Cov-2 spreading exponentially, it is needed to make people aware of Covid-19 symptoms and its cure. This website focuses on detecting Covid-19 virus through symptoms and guiding the people to further take an appointment with a doctor. It simplifies the task of people searching for hospitals with proper facilities and getting their treatment. Using some prior information such as cough, headache, fever, and loss of smell the site would estimate how prone the person towards Covid-19 is. Other factors which decide the effect of coronavirus are age, gender, and other chronic diseases if any. Based on all the symptoms and other factors, the website generates the probability of the user being affected by coronavirus by using Naive bayes algorithm. For people who are tested positive, medicines recommended will be purely based on symptoms and regular instructions given to a citizen in accordance with the government protocols. For their convenience, they can even search for hospitals in their location. Using Google map API and GPS a list of hospitals in the vicinity would be displayed. The user can book an appointment with any of the listed hospitals through our website and can also get tested for Covid-19. The reports will be maintained in the database in a sequential order which can be downloaded anytime for in case a person gets affected by coronavirus for more than one time, the history would be maintained. The person is considered to have recovered after receiving a negative Covid-19 report. The users can regularly visit the site for their check-up and schedule an appointment with a doctor. A massive number of people visit the site and avail the facilities to keep themselves safe. This makes the website a perfect tool for citizens to examine Covid-19 and undergo proper treatment with all the facilities available to ensure quick recovery.

8. FUTURE WORK

In the future public health is going to be handled in a digital manner. Viruses such as coronavirus may even cause more destruction in the upcoming years. Hence, to prepare humankind to face such a devastation, it is necessary to build up more facilities so that people do not depend on the conventional ways of treatment and go ahead with advanced technologies allowing users to view a precise report based on trained models and datasets. Therefore, the project seeks to include many more features subsequently. Those features may inlude

ambulance facilities, virtual meeting with a doctor and also ordering prescribed medicines. Time ahead it is possible that the world suffers from another pandemic which compels people to stay at home and delays in work and jobs are caused, in this situation the website proves to be very useful in assisting citizens and directing them to follow correct measurements. The team members would work on the required features to keep the website up to date and add facilities for the convenience of the users. Hence, The website would always function for a noble cause and serve humanity amongst people.

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